

1. To Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs.

To Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching

```
GNU nano 8.7.1          escape.c *
#include <stdio.h>

int main() {
    char input[100];

    printf("Enter a command: ");
    fgets(input, sizeof(input), stdin);

    printf("Input received: %s", input);

    return 0;
}
```

Output screen shot:

```
sabiya@LAPTOP-DOV34MDC:~$ nano escape.c
sabiya@LAPTOP-DOV34MDC:~$ gcc escape.c -o escape
sabiya@LAPTOP-DOV34MDC:~$ ./escape
Enter a command: hello\ worls
Input received: hello\ worls
sabiya@LAPTOP-DOV34MDC:~$ |
```

```
GNU nano 8.7.1 excute.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid;

    pid = fork();

    if (pid == 0) {
        printf("Child process created\n");
        execl("/bin/ls", "ls", NULL);

        printf("Execution failed\n");
    }
    else if (pid > 0) {
        wait(NULL);
        printf("Parent process completed\n");
    }
    else {
        printf("Fork failed\n");
    }

    return 0;
}
```

OUTPUT SCREENSHOT:

```
compilation terminated.
sabiya@LAPTOP-DOV34MDC:~$ gcc excute.c -o shell
sabiya@LAPTOP-DOV34MDC:~$ ./shell
Child process created
52Commands      WEEK1.7.c      mainloop.c     processconcepts1.c
FCFS.c          WEEK1.8.c      output.txt     processconcepts2.c
Fork.c          WEEK1.8.c.save parser          processconcepts3.c
Frok2.c         WEEK1.9.c      parser.c       processconcepts4.c
OSLab1.c        WEEK1.c        particalonetwo.c sabi.c
RoundRobyn.c    WEEK2.c        pipes          sabiya
SJFNonPreemptive.c dynamic.c      pipes.c        shell
WEEK1           escape         practicalonetwo.c signals.c
WEEK1-6.c       escape.c       practicaltwo.c singlequote
WEEK1.2.c       excute.c       priority.c     singlequote.c
WEEK1.3.c       filecopy      process.c      skill2.c
WEEK1.4.c       history.c     processconcept1.c tokenizer.c
WEEK1.5.c       input.txt     processconcept5.c
WEEK1.6.c       keyboard.c    processconcepts.c
Parent process completed
sabiya@LAPTOP-DOV34MDC:~$ |
```